

1 Solution — GIAM 8.4

Problem 3

1.1 Problem

Fill in the details of the proof that dominance is an ordering relation. (You may simply cite the C-B-S theorem in proving anti-symmetry.)

1.2 Setup

Section 8.4 defines:

If A and B are sets, we say " A dominates B " and write $|A| \geq |B|$ iff there is an injection $f : B \rightarrow A$.

(The textbook writes " $|A| > |B|$ " for dominance; the relation is *not* strict, so we use " $\geq$ " here to match standard notation.)

For $\geq$ to be an *ordering relation* on the class of cardinal numbers, three properties need verification:

Property	Statement
Reflexivity	$ A \geq A $ for every set A .
Transitivity	$ A \geq B $ and $ B \geq C $ imply $ A \geq C $.
Anti-symmetry	$ A \geq B $ and $ B \geq A $ imply $ A = B $.

Table 1.1

We take each in turn.

1.3 Reflexivity

We must exhibit an injection $A \rightarrow A$ for every set A . The *identity function* $\text{id}_A : A \rightarrow A$, defined by $\text{id}_A(x) = x$, does the job:

- *Well-defined*: every $x \in A$ maps to $x \in A$.
- *Injective*: if $\text{id}_A(x) = \text{id}_A(y)$, then by definition $x = y$.

So $|A| \geq |A|$. ■

1.4 Transitivity

Suppose $|A| \geq |B|$ and $|B| \geq |C|$. Unpacking:

- There is an injection $f : B \rightarrow A$.
- There is an injection $g : C \rightarrow B$.

We must produce an injection $C \rightarrow A$. The natural candidate is the *composition* $f \circ g : C \rightarrow A$, defined by $(f \circ g)(c) = f(g(c))$.

Well-defined. For any $c \in C$, the chain $g(c) \in B$ (since g has codomain B) and $f(g(c)) \in A$ (since f has codomain A) puts the output safely in A . So $f \circ g$ really is a function $C \rightarrow A$.

Injective. Suppose $(f \circ g)(c_1) = (f \circ g)(c_2)$, i.e.

$$f(g(c_1)) = f(g(c_2)).$$

Apply injectivity of f to conclude $g(c_1) = g(c_2)$, then apply injectivity of g to conclude $c_1 = c_2$. So $f \circ g$ is injective.

Therefore $|A| \geq |C|$. ■

1.5 Anti-symmetry

Suppose $|A| \geq |B|$ and $|B| \geq |A|$. Unpacking:

- There is an injection $f : B \rightarrow A$.
- There is an injection $g : A \rightarrow B$.

The **Cantor-Bernstein-Schröder theorem** (Section 8.4) states precisely that, in this situation, there is a *bijection* $A \rightarrow B$. By the definition of set equivalence,

$$|A| = |B|. \quad \blacksquare$$

1.6 Conclusion

Dominance is *reflexive* (identity), *transitive* (composition of injections), and *anti-symmetric* (C-B-S). It is therefore an ordering relation on the class of cardinal numbers.

1.7 Remark — Set vs. Cardinal

On the underlying class of *sets*, the dominance relation $\geq$ is reflexive and transitive but *not* anti-symmetric in the strict " $A = B$ " sense — two distinct sets like $\{1, 2\}$ and $\{a, b\}$ dominate each other but are not literally equal as sets, only *equivalent*.

So on sets, $\geq$ is a *preorder*; anti-symmetry only kicks in after we mod out by set equivalence. The quotient — cardinal numbers — is where dominance becomes a genuine partial order.

Object	Structure of $\geq$
All sets	Preorder (reflexive + transitive).
Cardinal numbers	Partial order (preorder + anti-symmetry, via C-B-S).

Table 1.2

This is the standard pattern in mathematics: a *preorder* + an *equivalence relation* whose quotient gives a *partial order*. The textbook's footnote — "the things really being compared are the cardinal numbers of sets" — gestures at exactly this construction.

1.8 Remark — Is It Also a Total Order?

A *total* (or *linear*) order would additionally require *trichotomy*: any two cardinals are comparable,

$$|A| \leq |B| \text{ or } |B| \leq |A|.$$

In ZFC, this is true — but the proof requires the Axiom of Choice. The argument: AC implies the *well-ordering theorem*, which lets us put any set in bijection with some ordinal; ordinals are totally ordered; cardinals (initial ordinals) inherit this order, so any two sets are comparable.

In *ZF without choice*, however, there can exist sets whose cardinals are genuinely incomparable — neither dominates the other. So:

- *ZFC*: dominance is a **total order** on cardinals.
- *ZF alone*: dominance is only a **partial order** on cardinals.

The textbook works inside ZFC throughout, and treats cardinals as totally ordered. Worth knowing that this is a non-trivial assumption.

1.9 Remark — The Strict Version $|A| < |B|$

The *strict* version of dominance is defined by

$$|A| < |B| \iff |A| \leq |B| \text{ and } |A| \neq |B|.$$

This is useful — Cantor’s theorem is phrased as $|A| < |\mathcal{P}(A)|$, and Problem 2 above placed $|\mathbb{R}| < |\mathcal{F}|$.

The strict version is *irreflexive* and *transitive* — a *strict partial order* — and is interdefinable with the non-strict one:

$$|A| \leq |B| \iff |A| < |B| \text{ or } |A| = |B|.$$

So once we have $\leq$ as a partial order, the corresponding facts about $<$ come for free.

1.10 Remark — Why C-B-S is the Hard Part

Reflexivity and transitivity are essentially trivial — they follow from the basic logic of functions (identity is an injection; composition preserves injectivity). The substantive content of “dominance is an ordering” lives entirely in anti-symmetry, which is the Cantor-Bernstein-Schröder theorem.

That asymmetry of difficulty reflects something genuine: producing two injections $f : A \rightarrow B$ and $g : B \rightarrow A$ is usually *much* easier than producing one explicit bijection $A \rightarrow B$. C-B-S says we don't have to — two injections going opposite ways are *enough* to certify equivalence.

This makes dominance an enormously useful comparison tool in cardinality arguments: to prove $|A| = |B|$, exhibit two ad-hoc injections rather than wrestle with one explicit bijection. A canonical example, mentioned in Section 8.4 itself: the unit interval $(0, 1)$ and the set of infinite binary strings F_2^∞ . A direct bijection is fiddly because of dual binary expansions ($0.0\overline{1} = 0.1\overline{0}$); two injections (one trivial, one via the “shift down by one position” trick from the Hilbert-Hotel discussion) are easy, and C-B-S finishes the job.